

Digital Input output (DIO)

* Digital component

↳ control (0,1)

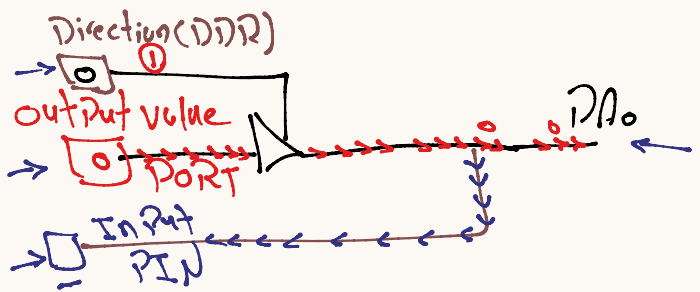
How circuit
Memory

How circuit

* overview

↳ Buffer

↳ tri state Buffer



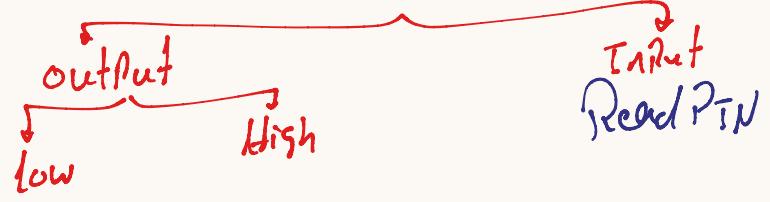
DDR → direction = 0 → Input

PIN → Input value → Reading

DDR → direction = 1 → output

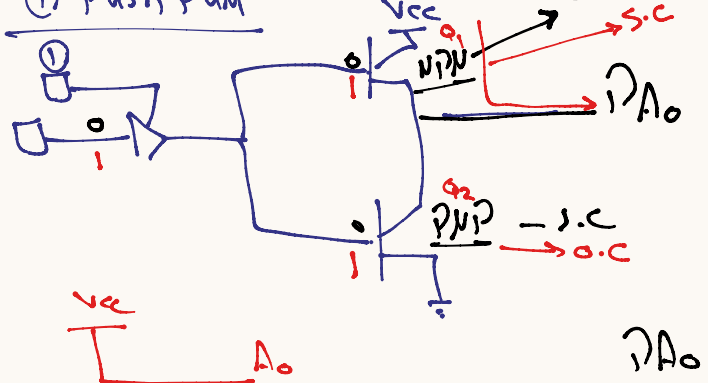
PORT → output value → 0

Direction



* output circuit type

① Push Pull



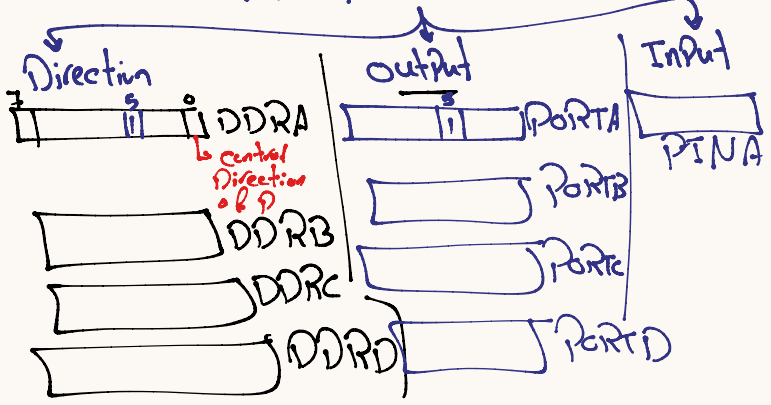
out	Q1	Q2	PA0
0	o.c	s.c	0V
1	s.c	o.s	5V

* Memory

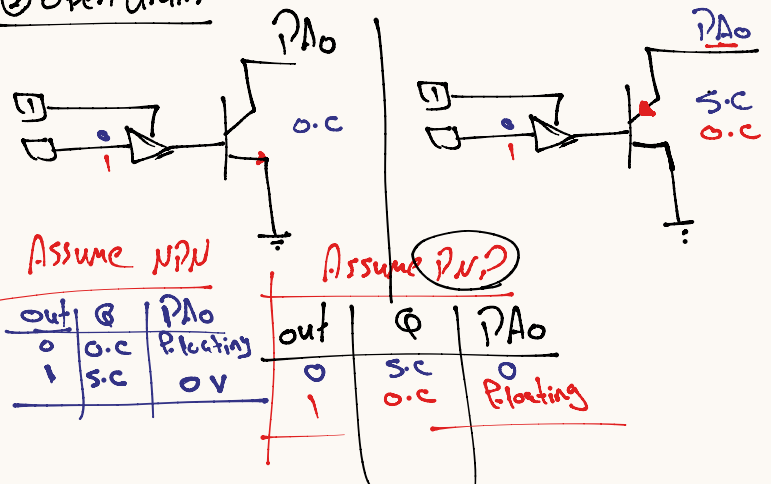
* to control on one pin you need 3bit from memory

* in Atmega32 you have 32 pin can be work as DIO

DIO Memory Reg



② open drain



Assume NPN

out	Q	PA0
0	o.c	floating
1	s.c	0V

Assume PNP

out	Q	PA0
0	s.c	o.c
1	o.c	floating

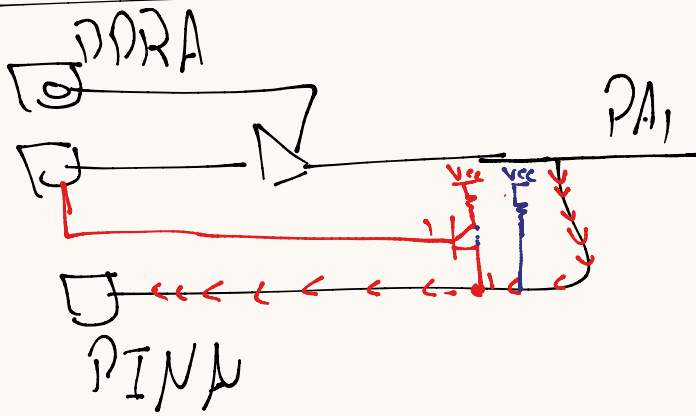
* in Atmega32

all pin work Push Pull

* only two pin I2C

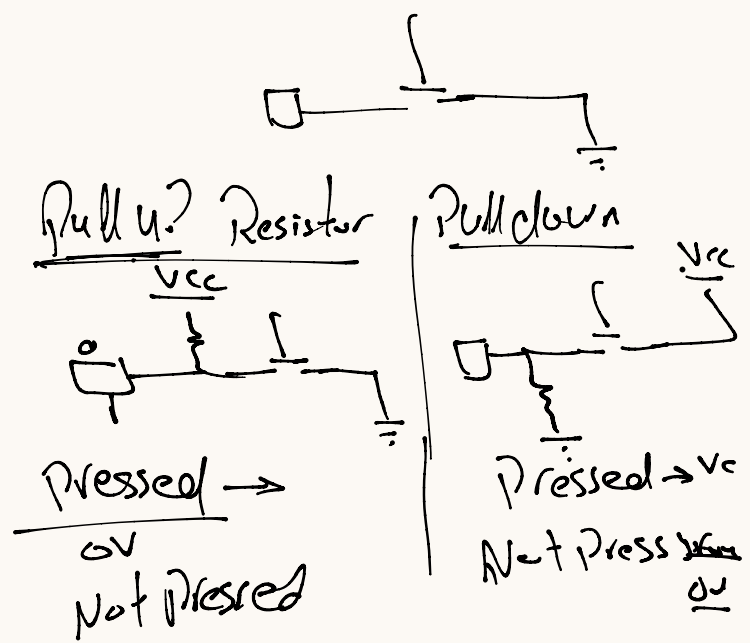
Can work push pull as DIO, Can work open drain I2C

* Internal Pull up

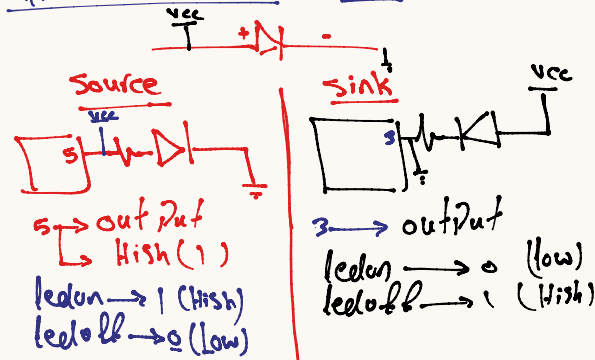


Enable
 ① DDR → 0
 ② PORT → 1

Disable



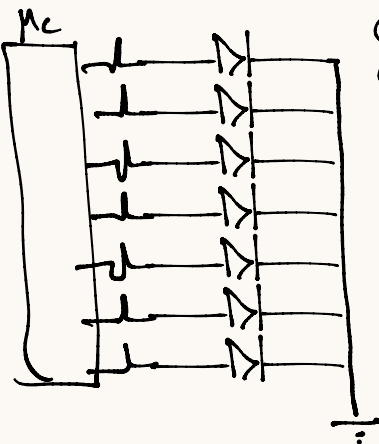
How Component → led



* Sink it's Better

Search pin from Mcu provide
 5V Max
 40mA Max
 Recommendation
 20:25 mA

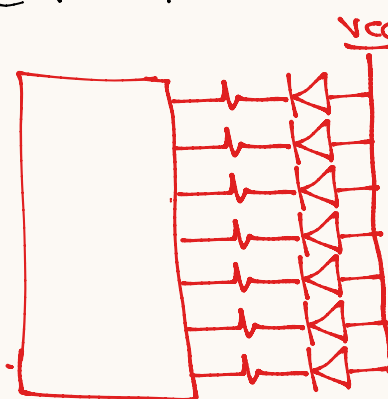
Connection 7 led → Source



Common ground
 Common Cathode

led on → 1 High
 led off → 0 Low

Connection 7 led Sink

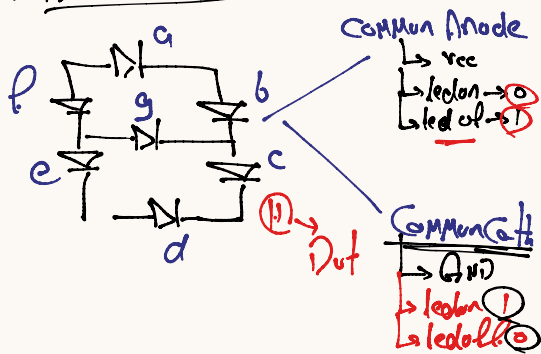


Common VCC
 Common Anode

led on → 0
 led off → 1

* 7. Segment

→ Display No



→ DDRC = 0xFF;

H 9 F e d c b a
7 6 5 4 3 2 1 0

Anode 1 0 1 1 0 0 0 0 → 0XB0

Cath 0 1 0 0 1 1 1 1 → 0X4F

a b c d e f g h
7 6 5 4 3 2 1 0

3 →
a
b
c
d

Switches →

→ Mechanical switch → Power human
→ electrical switch

Mechanical switches

Normal Button
save state

Push Button
→ signal

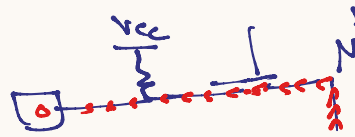


* → input

Connection

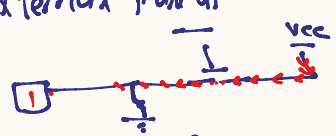
Pull up Resistor

Pull down Resistor



external Pull up

pressed = 0
Not pressed = 1



external Pull down

pressed = 1
Not pressed = 0

Must be Enable Internal Pullup

